

Understanding Enterprise Architecture and Framework

I. Research Questions

- A. **System Integration** is the process of connecting different applications, information systems and software so they can function together as a whole. Its main goal is to allow data to move easily between systems. By integrating systems, improve accuracy and increase operational efficiency. System integration also supports realtime data sharing which helps the admin to make better decisions.
- B. **System Architecture** is the conceptual model that defines the structure, behavior and views of a system. It provides a representation that shows how software, data, hardware, software, and users interact. System architecture ensures that all parts of a system work together to meet performance requirements. It also guides the developers and engineers during system development.
- C. **Enterprise Information Architecture (EIA)** is a framework that organizes organization data and information. It defines how information is stored, managed, collected and shared across the organization. EIA make sure that data is consistent, accessible and accurate to users. This architecture improves communication between departments.
- D. **Enterprise Architecture Planning (EAP)** is a structure used to define organization future planning and the guide to achieve it. It aligns the business goals with information systems and technology strategies. It identifies current systems, evaluates and recommends improvements. This process ensures that IT investments support long term projects.
- E. **Enterprise Application Integration (EAI)** is the process connecting an organization business, application, services, databases and other systems. It focuses on integrating software applications rather than entire systems. It allow organizations to continue using existing applications.

II. Compare and Contrast

1. Architecture and Design

Similarities

- Architecture and design both use planning before a system is built. They aim to ensure that a system meets the needs of the user, works properly and achieves its purpose.

Differences

- Architecture focuses on the entire structure and organization of a system in general. Design on the other hand focuses on detailed specifications of each component and how they will be implemented. Architecture provides the foundation of a system, while design provides the construction details.

2. Enterprise Architecture and Enterprise Engineering

Similarities

- Both enterprise architectures and enterprise engineering aim to improve the quality and how an organization operates.

Differences

- Enterprise architecture focuses on structuring and documenting the organization system, data and technology. Enterprise engineering is more on improving business processes and performance using engineering principles.

III. Zachman Framwork

	What (Data)	How (Function)	Where (Network)	Who (People)	When (Time)	Why (Motivation)
Planner						
Owner						
Designer						
Builder						
Subcontractor						
System						

Zachman Framework is used to organize and determine enterprise architecture artifacts. It ensures that all important parts of an organization like data, process, people and many others are documented. By using this framework, organization avoid missing important viewpoints. It also helps to align business needs with IT systems.

REFERENCES

Wikipedia contributors. (2024). System integration.

https://en.wikipedia.org/wiki/System_integration

Wikipedia contributors. (2024). Systems architecture.

https://en.wikipedia.org/wiki/Systems_architecture

Metaphacts. (2023). What is enterprise information architecture?

<https://metaphacts.com/what-is-enterprise-information-architecture>

ProductPlan. (2024). Enterprise architecture planning (EAP).

<https://www.productplan.com/glossary/enterprise-architecture-planning/>

TechTarget. (2024). Enterprise application integration (EAI).

<https://www.techtarget.com/searchapparchitecture/definition/EAI-enterprise-application-integration>

Wikipedia contributors. (2024). Zachman Framework.

https://en.wikipedia.org/wiki/Zachman_Framework